Multi-Backbone SED Ensembles with Snippet-Bank Matching for Indic Noise Event Detection

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Abstract—We describe our Track 1 system for the Datathon@IndoML 2026 noise event detection challenge. The task is to localise class-agnostic onset/offset pairs of noise events in Indic speech from the Vaani corpus, scored by event F1 plus segment Dice [10]. Our system combines three parts. (i) An ensemble of twelve frame-level sound event detectors built on diverse pretrained encoders (ATST-F, BEATs, WavLM-base+/large, w2v-BERT 2.0), each fused with a mel-CNN and a BiGRU emitting 24 presence and boundary tracks. (ii) A snippet bank: the organizers' synthetic mixtures reuse sample-exact noise snippets, which we isolate from validation pairs and detect with calibrated matched filtering. (iii) A routing decoder, tuned under test-mixture weighting, that merges bank and neural evidence per clip. Our final submission (Codabench 995227) reaches event F1 0.7711 and segment Dice 0.8232, a combined score of 1.5943 (rank 6), up from 1.2129 for our Phase 1 entry.

I. Training Data and Preprocessing

Vaani Noise Event Dataset [1] (90,637 segments, 154.6 h, seven noise categories).

- Gold (verified timestamps; 11,111 segments, 21.8 h) and Silver (unverified timestamps; 61,642 segments, 100.3 h) provide frame targets.
- Bronze (17,884 segments, 32.4 h; tags without timestamps) trains a multiple-instance (MIL) clip head.

Audio is staged once into a float16 blob with a JSONL manifest.

IndoML validation set (2,501 clips: 1,806 natural, 695 synthetic with clean references). It is split into five folds. Fold 4 (479 clips: 344 natural, 135 synthetic) is the monitor and selection hold-out. Natural clips of folds 0–3 are an extra training source for six of the twelve members; the other six never see validation audio.

Targets. Each clip yields 24 channels at a 20 ms frame rate: union presence, seven category presence tracks, and union plus per-category onset/offset tracks (Gaussian boundary targets).

Synthetic-mixture generator. The synthetic clips are built as noisy = clean + $\sum_i g_i$ snippet$_i$, with labels exact to the sample. We regenerate such mixtures on the fly, using event counts, gaps ($\geq$200 ms), gains and durations measured on the 695 validation synthetic clips.

- Noise: the validation snippet bank and, for widened members, isolated Gold event spans.
- Speech: validation clean references (held-out folds excluded) and Gold/Silver speech regions at least 50 ms from any event.

Batches and augmentation. Every batch mixes the sources in fixed proportions, for example Gold 0.40, Silver 0.27, Bronze 0.05 and synthetic 0.28. Crops are event-biased: 80% contain a whole event and 10% are hard negatives, with 10 s crops (4 s for the breath specialist). Augmentations are random gain, label-preserving whole-frame time shifts, label-union mixup within a tier, Filter-Augment [7] band gains, and optional Freq-MixStyle.

II. Proposed Method

A. Neural SED members

Each member fuses a pretrained encoder with a mel-CNN (512-dimensional projection, part dropout 0.1). A masked 2-layer BiGRU (hidden size 256) then outputs the 24 tracks, alongside a linear-softmax MIL head with seven clip logits.

Pretrained encoders:

- ATST-F strong and BEATs strong [2], [3], [4], from the PretrainedSED release v0.0.1;
- microsoft/wavlm-base-plus and wavlm-large [5];
- facebook/w2v-bert-2.0 [6].

The twelve members:

- m1 (ATST-F) and m4 (WavLM-base+ breath specialist: 4 s crops, human non-speech channels weighted $\times$3);
- m7 and m8 (ATST-F with seed 77, and with all-layer mixing);
- m1v, m1g and m6 (ATST-F, also trained on validation-natural clips; m1g continues on Gold data);
- m2 (BEATs), m3 (WavLM-large) and m5 (w2v-BERT), trained with widened mixtures that add Gold events;
- our Phase 1 WavLM-base+ detectors s9000 and s1600, which use learned layer weights, a BiGRU, 1+7+64 tracks, boundary heatmaps and attention-MIL.

Losses.

- presence: BCE or focal loss per track, plus a soft-Dice loss on the union track;
- boundaries: RED-style focal loss on the onset/offset tracks;
- Bronze clips: MIL loss only;
- tier weights (presence / boundary): Gold 1.0/1.0, Silver 0.5/0.1, synthetic 1.0/1.0, validation natural 1.5/1.5. Silver background frames are weighted $\times$0.5.

Training.

- Stage 1 trains the new modules with the encoder frozen (1,000–1,500 steps, learning rate 10^{-3}).
- Stage 2 unfreezes the encoder (learning rate $1–2{\cdot}10^{-5}$, layer-wise decay 0.85–0.9).
- AdamW, bf16, gradient clipping, and EMA weights [8] throughout.
- A cosine schedule is fitted to a wall-clock deadline from measured step times.
- A fold-4 monitor keeps the two best checkpoints (student or EMA).

Inference uses 10 s windows with a 5 s hop, triangular overlap weights (floor 0.3), and time-shift TTA over $\{0, \pm 20\}$ ms.

B. Snippet bank

Subtracting clean from noisy audio at the labelled spans of the 695 validation synthetic pairs isolates 2,213 events, which deduplicate to 206 distinct snippets.

Candidate generation. For every snippet, normalised cross-correlation (NCC) is computed against each clip via FFT. Up to three peaks with NCC ≥ 0.35 are kept per snippet. Edge sub-templates (the first or last 250 ms) catch events that the clip boundary truncates.

Scoring. Each candidate is described by its NCC, a chunk-lag consistency score (whether every 50 ms chunk peaks at the same lag, a property of sample-exact copies but not of natural look-alikes), the implied SNR, the snippet duration, and an edge flag. A logistic model calibrated leave-fold-out maps these features to a match probability, and greedy non-maximum suppression produces the bank events.

C. Fusion, routing and decoding

Fusion. The member caches are fused in the logit domain. The final configuration (clean_plus_a30) gives 70% of the fusion mass to the six members never trained on validation audio and 5% to each of the other six.

Routing. A calibrated synthetic score sends each clip down one of three paths:

- bank: bank events above 0.56 confidence, merged with neural events above 0.64 confidence (margin 0.18, gap 200 ms, at most five events);
- natural: a category-aware hysteresis decoder (thresholds 0.85/0.61, median filter of 5 frames, minimum 40 ms, merge gap 160 ms, boundary padding $-20/-30$ ms) that splits blobs whose category tracks disagree, so overlapping events of different types become separate events;
- prior: a fallback. No clip is ever left empty.

Tuning. All thresholds and fusion weights are searched with Optuna TPE [9] on validation, with synthetic clips weighted $\times 1.57$ to match the test mixture of 2,080 synthetic and 3,437 natural clips. The tuner reports a 5-fold CV estimate next to the full-validation score.

TABLE I
Combined score (event F1 + segment Dice) on validation and on the Codabench test set. Ensembles with validation-trained members include training folds in the 2,501-clip column; fold 4 is held out.

ID	Configuration	Val. (2,501)	Fold 4 (479)	Test
938975	s9000+s1600 (Phase 1)	1.134	–	1.2129
995013	interim, clean members	1.468	1.466	1.5803
995109	all 12, fold-4 tuned, test-mix	1.541	1.512	1.5854
995114	all 12, fold-4 tuned	1.541	1.512	1.5857
995138	clean members, test-mix	1.482	1.465	1.5875
995227	clean + 30% others	1.533	1.480	1.5943

III. Results and Conclusion

Table I lists our Phase 1 entry and every Phase 2 submission.

- Final entry (995227): event F1 0.7711 and segment Dice 0.8232 on the 5,517 test clips, combined 1.5943. It routes 1,974 clips through the bank (the test set has 2,080 synthetic clips), 3,482 through the natural decoder and 61 through the prior, emitting 13,268 events.
- The bank transfers to the test set. Event F1 rose from 0.47 for our Phase 1 entry to 0.77.
- Members trained only on non-validation audio generalise best. clean_full_tm scores lower than final_f4_tm on validation but higher on the test set. Re-adding the validation-trained members at 30% total weight gave the best test score.
- Category-aware decoding matters for natural clips. About 30% of consecutive natural event pairs overlap, and splitting fused blobs by category track recovers them as separate events.

Novelty. Our contributions are (i) exploiting the reuse structure of the synthetic mixtures with calibrated, consistency-checked matched filtering; (ii) a generator that reproduces the organizers' mixtures with labels exact to the sample, used for training; (iii) a heterogeneous ensemble spanning five pretrained encoders, fused in logits under a held-out-aware weighting; and (iv) test-mixture-weighted decoder tuning with routing.

Conclusion. Combining heterogeneous pretrained SED members with an explicit model of how the synthetic data was built gives a strong and well-calibrated detector. Natural clips remain the harder half; better boundary modelling of overlapping natural events is the main avenue for improvement.

References

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